

# Kaichen Xu

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## EDUCATION

### Emory University

Sep 2025 – Present

**Major:** Master of Science in Computer Science      **GPA:** 3.90/4.00

### Zhongnan University of Economics and Law (ZUEL)

Sep 2020 – Jun 2024

**Major:** Bachelor of Science in Statistics      **GPA:** 3.85/4.00 (92.21/100)

## RESEARCH INTERESTS

Generative Modeling; Computational Biology & Bioinformatics; Gene Regulatory Network Inference

## PUBLICATIONS

\*: equal contribution

- [1] **Kaichen Xu**, Yan Lu, and et al. Detecting and Dissecting Anomalous Anatomic Regions in Spatial Transcriptomics with STANDS. *Nature Communications*, 2024.
- [2] **Kaichen Xu**, Yihang Du, and et al. Causality-Induced Positional Encoding for Transformer-Based Representation Learning of Non-Sequential Features. In *Proceedings of the International Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
- [3] **Kaichen Xu**, Mianpeng Liu, and et al. Concord: Building Consensus Representations for Single Cells with Collaborative Random Projection. In *Proceedings of the SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2026.
- [4] **Kaichen Xu\***, Qilong Wu\*, and et al. Multimodal Anomalous Tissue Region Detection Enhanced with Spatial Transcriptomics. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2025.
- [5] **Kaichen Xu\***, Yueyang Ding\*, and et al. Domain Adaptive and Fine-grained Anomaly Detection for Single-cell Sequencing Data and Beyond. In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2024.
- [6] **Kaichen Xu\***, Kainan Liu\*, and et al. Detecting and Subtyping Anomalous Single Cells with M2ASDA. In *ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM BCB)*, 2024.
- [7] Tao Deng, Mengqian Huang, **Kaichen Xu**, and et al. LEGEND: An Integrative Algorithm for Identifying Co-expressed and Cofunctional Genes in Multimodal Transcriptomic Sequencing Data. *Genomics Proteomics & Bioinformatics*, 2025.

## RESEARCH EXPERIENCES

### Causality-Induced Positional Encoding for Transformer-Based Representation Learning of Non-Sequential Features

- Investigated the theoretical relationship between sequential ordering and causal dependency structures.
- Proposed CAPE (Causality-Aware Positional Encoding), a framework integrating causal discovery with transformer architectures to generalize positional encoding beyond sequential inputs.
- Enabled attention mechanisms to capture causal strength and structural dependencies among unordered features, improving representation learning in tabular and structured datasets.
- Conducted theoretical analysis and large-scale empirical evaluations demonstrating significant improvements over standard positional encoding strategies.
- Published at [NeurIPS 2025](#).

### Detecting and Dissecting Anomalous Anatomic Regions in Spatial Transcriptomics with STANDS

- Studied disease heterogeneity and microenvironmental variation in spatial transcriptomics data.
- Identified key challenges in population-level versus patient-specific pathological patterns affecting disease characterization.
- Proposed STANDS, a multimodal deep learning framework based on Generative Adversarial Networks integrating spatial gene expression profiles with histopathology (H&E) imaging.
- Achieved state-of-the-art performance in detecting pathological tissue regions across multiple spatial transcriptomics datasets.
- Published in [Nature Communications](#) (undergraduate first author).

## Domain Adaptive and Fine-grained Anomaly Detection for Single-cell Sequencing Data and Beyond

- Investigated the impact of domain shifts across multi-sample biological datasets on anomaly detection performance.
- Proposed ACSleuth, a novel anomaly detection framework inspired by Maximum Mean Discrepancy (MMD) and Ramanujan's Master Theorem.
- Developed algorithms capable of detecting rare cell types and subtypes in large-scale single-cell sequencing datasets.
- Extensive experiments demonstrate improved robustness to domain shifts and superior detection accuracy.
- Published at [IJCAI 2024](#) (undergraduate first author).

## WORK EXPERIENCES

### Emory University

Atlanta, GA

Research Assistant (Advisor: [Prof. Victor Corces](#))

Mar 2026 – Present

- Conduct research on alternative splicing and RNA regulation using multimodal single-cell genomics data.
- Investigate how sequence context and chromatin accessibility jointly regulate RNA processing, providing a predictive framework for alternative splicing across cell types.
- Led a research project as co-first author, preparing the manuscript for submission to Nature Genetics.

### The Chinese University of Hong Kong (Shenzhen)

Shenzhen, China

Research Assistant (Advisor: [Prof. Jin Liu](#))

Jun 2025 – Dec 2025

- Conducted research on gene regulatory network (GRN) inference from single-cell transcriptomic data, aiming to uncover regulatory interactions among genes in complex biological systems.
- Developed scalable and interpretable single cell foundation models for inferring gene-gene regulatory relationships.
- Led a research project as first author; the resulting manuscript is currently under review at Nature Biomedical Engineering.

### Zhongnan University of Economics and Law

Wuhan, China

Research Assistant (Advisor: [Prof. Xiaobo Sun](#))

Mar 2023 – Dec 2025

- Conducted research on anomaly detection, multimodal biological data integration, and deep learning theory.
- Mentored junior undergraduate researchers in machine learning and computational biology, providing guidance on research design, experimentation, and paper writing; mentees subsequently achieved publications in major AI conferences.
- Research outcomes resulted in first-author publications in Nature Communications, IJCAI, AAAI, and NeurIPS.

## LEADERSHIP & MENTORSHIP

### Organizer and Speaker — Machine Learning Academic Seminar Group

- Founded a weekly online seminar on machine learning research, engaging over 30 participants from multiple universities.
- Delivered monthly presentations on recent ML papers, facilitating discussions on emerging research directions.
- Mentored participants in research methodology and project development; members later produced 10+ conference papers in AI and computational biology venues.

### Server Administrator and Research Mentor — Sunlab

- Managed the lab's servers and computing infrastructure, ensuring reliable access for large-scale machine learning experiments.
- Conducted 30+ interviews for prospective lab members and contributed to research group recruitment.
- Mentored junior researchers in machine learning and experimental design; one mentee published a paper at IJCAI 2025 as an undergraduate.

## INVITED TALKS & PRESENTATIONS

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|----------------------------------------------------------------------------------------------------------------|----------|
| Oral Presentation at ACM SIGKDD Conference on Knowledge Discovery and Data Mining (BioKDD workshop)            | Aug 2026 |
| Oral Academic Seminar for introducing our work STANDS, Peking University                                       | Dec 2024 |
| Oral Presentation at ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM BCB) | Nov 2024 |
| Oral Presentation at International Joint Conference on Artificial Intelligence (IJCAI)                         | Aug 2024 |

## TECHNICAL SKILLS

**Programming Languages:** Python (PyTorch, TensorFlow, Pandas, NumPy, Scanpy, AnnData, etc.), R, C/C++, PHP

**Technologies and Tools:** Latex, Linux, GitHub, MATLAB, MySQL, Markdown, HTML/CSS, PS, AI